

# LLM Validation Log

**Course:** FINTECH 590.05 Financial Decision Systems

**Deliverable:** Defense 1, Risk Committee (Lecture 6)

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**Artifacts covered:** Fintech590\_Defense1\_RiskConsole\_SlideDeck.pbix (slide deck version of the console, Parts 1 to 3, with its PDF export) and the Institutional Risk Memo

**AI tools used:** Claude (Anthropic) for the console and initial log (rows 1-17); Codex (OpenAI) for the final memo, raw-source audit, and formula workbook (rows 18-25); Claude (Anthropic) for the slide deck version of the console (rows 26-32).

## How verification was done

Nothing below was accepted because it looked right. Three independent checks were used:

- 1. Recomputation from the source file.** The FFIEC FR Y-9C bulk file (data/BHCF20260630.ZIP, quarter ending 2026-06-30) was parsed separately in Python, outside Power BI, and every metric and every stress path was recalculated from the raw MDRM fields. Output saved as Defense 1/stress\_paths\_Q0\_Q9.csv (150 rows: 5 banks x 3 scenarios x 10 quarters).
- 2. Cross-check against regulator reported values.** Computed Tier 1 ratios were compared against BHCA7206, the Tier 1 ratio each holding company reports on Schedule HC-R, and against the banks' published quarterly disclosures.
- 3. Model level data quality checks.** Row count, distinct RSSD count and a name mismatch count are measures on the Bank Detail page, so the checks run on every refresh rather than once.

A final comparison was run on 2026-09-29: a DAX query returned Q0 and Q9 Tier 1, Q9 liquidity gap, Q9 securities loss as a share of Tier 1, Q9 runoff dollars and total securities for all 5 banks under all 3 scenarios. All 15 rows matched the Python recomputation exactly at 2 decimal places.

## Log

**Entry 01 — Claude: build a Power Query query that reads the FFIEC FR Y-9C bulk ZIP, filters to our 5 RSSD IDs and pulls the fields behind the 5 required metrics**

**Produced.** Metrics query in M: unzips the file in memory, promotes headers, filters on RSSD9001, selects and renames 20 MDRM fields, adds total deposits, rate sensitive liabilities and securities

**Verified.** Parsed the same file in Python and compared all raw fields for all 5 banks, value by value. Every field matched. Tier 1 ratios also matched the banks' own reported BHCA7206 (for example JPMorgan 15.13% computed vs 15.1339 reported)

**Correction.** None to the logic. The source was switched from a teammate's Excel export to the official FFIEC ZIP so the numbers trace to a primary source

### **Entry 02 — Claude: which MDRM fields correspond to each required metric**

**Produced.** Field map: Tier 1 capital BHCA8274, RWA BHCAA223, average assets for leverage BHCAA224, loans BHCK2122, deposits BHDM6631 + BHDM6636 + BHFN6631 + BHFN6636, past due BHCK1407, nonaccrual BHCK1403, rate sensitive assets BHCK3197, repricing liabilities BHCK3296 + BHCK3298 + BHCK3408 + BHCK3409, total assets BHCK2170, securities BHCK1754 + BHCK1773

**Verified.** Each code was located in the FFIEC file header and its value compared with the schedule it should come from. Past due and nonaccrual codes were confirmed by matching the values already in the team workbook

**Correction.** None. Two codes we assumed at first (BHCK5525, BHCK5526) do not exist in the Y-9C file; corrected to BHCK1407 and BHCK1403 before use

### **Entry 03 — Claude: write the DAX for the five required metrics from the raw fields**

**Produced.** Tier 1 Ratio, Leverage Ratio, LTD Ratio, NPL Ratio, Liquidity Gap 12M, each a DIVIDE over SUM of raw columns

**Verified.** Recomputed all five for all five banks in Python and compared. Exact match

**Correction.** None

### **Entry 04 — Claude: write a formula driven conditional color measure for the Tier 1 cards**

**Produced.** Tier1\_Ratio\_Color = SWITCH(TRUE(), [Tier 1 Ratio] >= 0.10, green, >= 0.08, amber, red)

**Verified.** Checked each threshold branch by substituting test ratios (7.9%, 8.0%, 9.9%, 10.0%) and confirming the returned value, then confirmed the card recolors when the underlying capital figure is edited and the model refreshed

**Correction.** The lecture example returned the words "Green" and "Amber". "Amber" is not a valid color name and Power BI ignores it, so the measure returns hex codes (#2E7D32, #F9A825, #C62828) instead

### **Entry 05 — Claude: create the measures in the model**

**Produced.** Measure names Tier1\_Ratio, NPL\_Ratio and similar

**Verified.** Attempted to write them to the model; Power BI rejected them

**Correction.** Names collided with existing column names from the source data. Renamed to spaced names (Tier 1 Ratio, NPL Ratio, Liquidity Gap 12M)

### **Entry 06 — Claude: build a Power Query table holding the three CCAR scenario paths from the assignment table**

**Produced.** Scenario\_Path query: 30 rows (3 scenarios x Q0 to Q9) with cumulative Tier 1 drift in points, cumulative deposit runoff percent and securities haircut percent

**Verified.** Recomputed the same 30 rows in Python from the assignment text and compared. Match. Spot checked the end points against the brief: adverse runoff 5% by Q4, severely adverse 15% by Q2, haircuts 3% and 10%

**Correction.** None to the arithmetic, but see row 7 for the interpretation decision

### **Entry 07 — Claude: flagged that the scenario table is internally inconsistent**

**Produced.** The per quarter Tier 1 rules sum to minus 1.0 pt (adverse) and minus 1.3 pt (severely adverse) by Q9, while the brief quotes about minus 1.6 and minus 3.1

**Verified.** Added the stated quarterly deltas by hand for both scenarios and confirmed the totals do not reconcile with the quoted figures

**Correction.** Team decision: follow the explicit per quarter rules, because each step then traces to a stated rule. The discrepancy is disclosed in the memo and the console footnote

### **Entry 08 — Claude: write the stressed metric measures**

**Produced.** Tier 1 Ratio Stressed (starting ratio plus cumulative drift), Deposit Runoff Amount, Deposits Stressed, Liquidity Gap 12M Stressed, Securities MTM Loss, Securities Loss Pct of Tier 1

**Verified.** All recomputed in Python for 150 bank, scenario and quarter combinations and compared against a DAX query pulling the same values out of the model. Exact match on all rows

**Correction.** None

### **Entry 09 — Claude: how should deposit runoff affect the liquidity gap**

**Produced.** Runoff is replaced by wholesale and FHLB borrowings, which reprice within 12 months, so the runoff amount is added to rate sensitive liabilities

**Verified.** Checked against the scenario table in the brief, which states the replacement funding explicitly. Verified the direction and size by hand for JPMorgan: gap falls from 29.7% to 21.6% under severely adverse, consistent with adding 15% of deposits to repricing liabilities

**Correction.** None. Stated as an assumption in the memo, since the behavioral path could differ

### **Entry 10 — Claude: should the securities haircut also reduce Tier 1 capital**

**Produced.** Reported the haircut as a mark to market loss and as a share of Tier 1, without deducting it from the Tier 1 ratio

**Verified.** Reasoned against the brief: the scenario prescribes the Tier 1 path separately, so deducting the haircut as well would double count the capital hit

**Correction.** None. Disclosed in the memo and on the console

### **Entry 11 — Claude: write a dynamic action title for the console**

**Produced.** Stress Action Title, which names the bank with the lowest Q9 Tier 1 and the bank with the lowest Q9 liquidity gap for the selected scenario

**Verified.** Switched the scenario slicer through all three settings and checked the named banks and values against the recomputed table each time

**Correction.** None

### **Entry 12 — Claude: reported that two RWA values looked hand entered (JPMorgan 2,132,427,745 and Wells Fargo 1,342,324,680)**

**Produced.** A suspicion that the figures had been derived rather than pulled

**Verified.** Looked both values up directly in the FFIEC file under BHCAA223. Both are exactly as reported by the banks

**Correction.** None needed. The suspicion was wrong and the workbook values were correct

**Entry 13 — Claude: initially stated the workbook values had been multiplied by 1000 and were not in original FFIEC units**

**Produced.** An incorrect claim about the units

**Verified.** Compared the workbook figures against the FFIEC file: both are in thousands of dollars, as filed

**Correction.** Claim retracted. Units are thousands throughout, and the README says so

**Entry 14 — Claude: pull Call Report RC-R for each holding company's lead bank through the FFIEC CDR API**

**Produced.** CDR\_Bank\_RCR query, authenticated against the CDR Public Data Distribution REST API

**Verified.** Compared bank level Tier 1 ratios against another team's independently built Call Report dashboard: Goldman Sachs Bank 13.5%, JPMorgan Chase Bank 15.0%, Citibank 14.9%, Bank of America 12.1%, Wells Fargo Bank 12.1%. All five matched

**Correction.** None

**Entry 15 — Claude: numbers quoted in the exposure note and memo (JPMorgan 12 month gap 29.7%, peer range 36.3% to 39.5%, repricing liabilities 16.4% of assets versus 8.8% to 11.6%, Tier 1 15.1%, leverage 6.6%)**

**Produced.** The figures used in the written deliverables

**Verified.** Each number was recomputed in Python from the raw FFIEC fields and matched against the value the dashboard displays

**Correction.** None

**Entry 16 — Claude: write a measure that explains the securities chart under each scenario**

**Produced.** Securities Chart Subtitle: returns a no haircut message when the haircut is 0% (Baseline), otherwise states the haircut applied

**Verified.** Ran a DAX query returning the subtitle for each of the three scenarios and checked each string; then switched the report slicer through all three and confirmed the chart subtitle changed accordingly

**Correction.** None

**Entry 17 — Claude: built the Tier 1 stress path line chart**

**Produced.** Line chart of Tier 1 Ratio Stressed by quarter and bank

**Verified.** Visual review against the liquidity chart and the recomputed path

**Correction.** The axis was sorted by the Tier 1 value, descending, so the path ran Q9 back to Q0. Changed the sort to quarter, ascending. A leftover duplicate of the chart, still sorted the old way, was also found hidden behind the liquidity chart and deleted. No numbers changed

## **Memo and independent-audit entries**

Rows 1-17 above document the teammate's Claude-assisted Power Query, DAX, dashboard, and earlier checks. Rows 18-25 document the Codex-assisted memo and separate audit. The audit was performed from the FFIEC ZIP and the written assignment rules, without reading the Power BI DAX.

### **Entry 18 — Codex: identify the most exposed bank from the Q0 FFIEC fields and team dashboard**

**Produced.** Memo figures: JPMorgan 15.1% Tier 1, 29.7% repricing gap; peer gaps 36.3%-39.5%; JPMorgan one-year repricing liabilities 16.4% of assets versus 8.8%-11.6% for peers

**Verified.** Re-extracted the five RSSD records from BHCF20260630.ZIP; calculated Tier 1 as BHCA8274/BHCAA223, gap as (BHCK3197 - BHCK3296 - BHCK3298 - BHCK3408 - BHCK3409)/BHCK2170, and liability shares directly from those fields. The Source tab retains the raw inputs.

**Correction.** Called the gap a repricing proxy rather than immediately available cash.

### **Entry 19 — Codex: quantify both required stress impacts for JPMorgan**

**Produced.** Adverse Q9: Tier 1 14.1%, gap 27.0%, changes -1.0 and -2.7 points; Severe Q9: 13.8%, 21.6%, changes -1.3 and -8.1 points

**Verified.** Applied the brief's quarterly Tier 1 steps and 5%/15% deposit runoff to the Q0 source fields; replacement funding is added to repricing liabilities. Independent Python results and the formula workbook (Summary, JPMorgan row; Paths, Q9 rows) agree with the team CSV at its reported precision.

**Correction.** Kept both the capital and liquidity effects, and described the scenarios as classroom stress paths rather than official CCAR results.

### **Entry 20 — Codex: check whether any bank's capital path is weaker than JPMorgan's**

**Produced.** Wells Fargo Severe Tier 1 reaches 9.6% at Q4 and 10.1% at Q9, the lowest Q9 capital ratio

**Verified.** Starting Wells ratio from raw BHCA8274/BHCAA223; added -0.6, -0.5, -0.4, -0.3 points through Q4, then +0.1 point per quarter through Q9. Checked workbook Summary row 10 and the team CSV Q4/Q9.

**Correction.** Added the Q4 trough; a Q9-only description would hide the lower interim capital point.

### **Entry 21 — Codex: quantify the securities shock without double-counting Tier 1**

**Produced.** JPMorgan Severe haircut exposure equals 24.9% of starting Tier 1 capital

**Verified.** Raw HTM+AFS securities are BHCK1754+BHCK1773 = 804,564,000 (USD thousands); 10% loss / BHCA8274 capital 322,720,000 = 24.9307%. Checked against the team CSV and Paths workbook result.

**Correction.** Explicitly stated that this exposure is separate from the prescribed Tier 1 path and is not deducted again.

### **Entry 22 — Codex: transcribe the assignment's scenario parameters and reconcile its capital totals**

**Produced.** Adverse 5% runoff by Q4 and 3% securities haircut; Severe 15% runoff by Q2 and 10% haircut. Quarterly Tier 1 steps sum to -1.0 and -1.3 points by Q9, versus the brief's approximate -1.6 and -3.1 point headlines.

**Verified.** Checked the scenario table in the Defense 1 brief; summed the explicit steps in workbook Assumptions H5:H6 and separately in Python. The workbook phases the haircut to the Q4/Q2 endpoint as an identified team assumption.

**Correction.** Followed the detailed quarterly rules, matching the teammate's console and CSV, and disclosed the inconsistency in the memo.

### Entry 23 — Codex: draft a committee recommendation and change conditions

**Produced.** Targeted JPMorgan funding review; separate Wells Fargo capital watch; escalation or de-escalation conditions

**Verified.** Checked that JPMorgan is the lowest gap in Q0, Adverse Q9 and Severe Q9, while Wells is the lowest Severe Q9 Tier 1 and dips further at Q4. The recommendation is a judgment based on the defined proxy, not a claim of imminent failure.

**Correction.** Replaced any implication that the gap is a cash reserve or that the class scenario is a regulatory CCAR forecast.

### Entry 24 — Codex: rebuild the stress paths in an independent Excel workbook

**Produced.** FINTECH590\_Independent\_Stress\_Check.xlsx with Source, Assumptions, Paths, and Summary tabs; 150 formula-driven rows

**Verified.** Raw FFIEC fields were extracted independently of Power BI. A separate Python calculation checked all six numeric CSV columns for all 150 rows within reporting precision; the workbook checks Tier 1 and gap for all 150 rows (150 PASS). The largest ratio difference before rounding was under 0.0000005.

**Correction.** No numeric discrepancies found. The workbook is AI-authored; the verification method is independent computation, not a claim of manual student entry.

### Entry 25 — Codex: turn the memo into a two-page submission file

**Produced.** FINTECH590\_Institutional\_Risk\_Memo.pdf and editable Markdown source

**Verified.** Compared every quantitative sentence with the raw-source audit and the workbook; reviewed both rendered PDF pages for legibility and completeness. The memo contains the four required elements: recommendation, capital/liquidity evidence, two scenario impacts, and change conditions.

**Correction.** Kept the recommendation conditional and included the assignment's Tier 1 inconsistency and model limits.

## Slide deck version entries

The slide deck version reuses every query and measure logged in rows 1 to 17 unchanged. Rows 26 to 32 cover the formulas added for the slide deck criteria (fan chart; Executive, Analyst and Regulator pages). Verification sources: Defense 1/stress\_paths\_Q0\_Q9.csv and the FFIEC NIC FR Y-9C bulk files for 2024-09-30 to 2026-06-30.

### Entry 26 — Claude: pull 8 quarters of reported Tier 1 history for the fan chart's actual series

**Produced.** data/tier1\_history\_Y9C.csv: 40 rows (5 banks x 2024-09-30 to 2026-06-30) of BHCA8274 and BHCAA223 from the FFIEC NIC bulk files BHCF20240930 to BHCF20260630, plus the reported BHCA7206

**Verified.** Recomputed capital / RWA for every row and compared with the reported BHCA7206: largest difference 0.00005 percentage points. The 2026-06-30 rows equal the Metrics query values

**Correction.** The first version of the file had an unquoted comma in the source column, which shifted fields; rewritten with 6 fields per row before loading

### Entry 27 — Claude: build a single date axis for actual history plus projections

**Produced.** FanDates query: 17 quarter ends from 2024 Q3 to 2028 Q3, q\_index -7 to 9, with Q0 = 2026-06-30

**Verified.** Checked every label against its quarter end date (for example q\_index 4 = 2027-06-30 = 2027 Q2) and confirmed the text labels sort in date order

**Correction.** None

### Entry 28 — Claude: write the fan chart measures

**Produced.** Fan Actual, Fan Baseline, Fan Severely Adverse, Fan Breach (only returns a value below 8%), Sev Trough Tier1, Sev Trough Q

**Verified.** DAX query of all 17 quarters for Wells Fargo compared with stress\_paths\_Q0\_Q9.csv: severely adverse trough 9.58% at projected quarter 4 in both, Q9 10.08% in both. Troughs for all five banks match the CSV ranking (WFC 9.58, BAC 10.84, GS 12.67, C 12.88, JPM 13.33)

**Correction.** Tooltips first showed decimals (0.114); set the measures to percentage format

### Entry 29 — Claude: write the data driven action title for the fan chart

**Produced.** Fan Title: names the bank with the lowest severely adverse trough and either the breach quarter or, if no breach, how close it gets

**Verified.** Returned "No bank breaches 8% under Severely Adverse: Wells Fargo comes closest, bottoming at 9.6% in 2027 Q2 (projected quarter 4 of 9)". Checked bank, value and quarter against the CSV

**Correction.** Earlier chat text called the trough "Q4 2027"; corrected to projected quarter 4, which is calendar 2027 Q2

### Entry 30 — Claude: make the breach series red only below 8%

**Produced.** Breach series color set to red; measure returns blank at or above 8%

**Verified.** Temporarily raised the threshold to 12% so the series drew, set its color, then restored 8% and confirmed the series is empty for every bank, since no trough is below 8%

**Correction.** None

### Entry 31 — Claude: write data driven action titles for the Analyst page panels

**Produced.** Analyst Tier1 Title, Analyst Liquidity Title, Analyst Securities Title

**Verified.** DAX query of all three titles under each scenario, checked against the CSV (for example severely adverse: WFC 9.6% in Q4, JPM gap 21.6% by Q2, BAC securities loss 37.7% of Tier 1)

**Correction.** Baseline first read "bottoming at 11.4% in Q0"; reworded so a scenario with no decline says the ratio never dips below its starting value

### Entry 32 — Claude: build the Regulator page

**Produced.** Liquidity gap inputs table with the MDRM code in every column header (BHCK3197, 3296, 3298, 3408, 3409, 2170), RSSD traceability table, neutral fan chart title

**Verified.** Each column was compared with the Metrics query field it comes from, and the headers against the FR Y-9C HC-H item codes

**Correction.** None